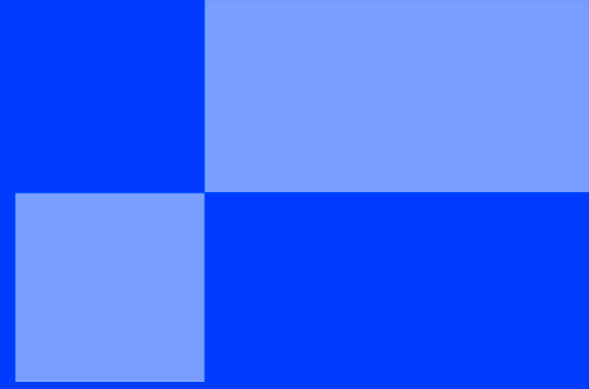
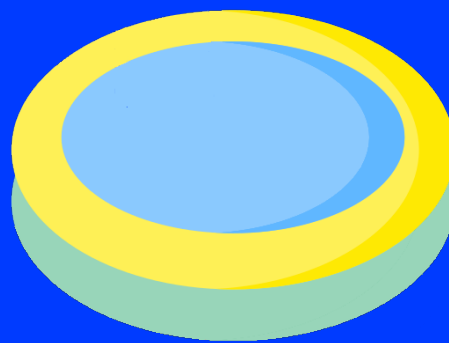




PREPARATION BOOTCAMP

< CONTROL FLOW />

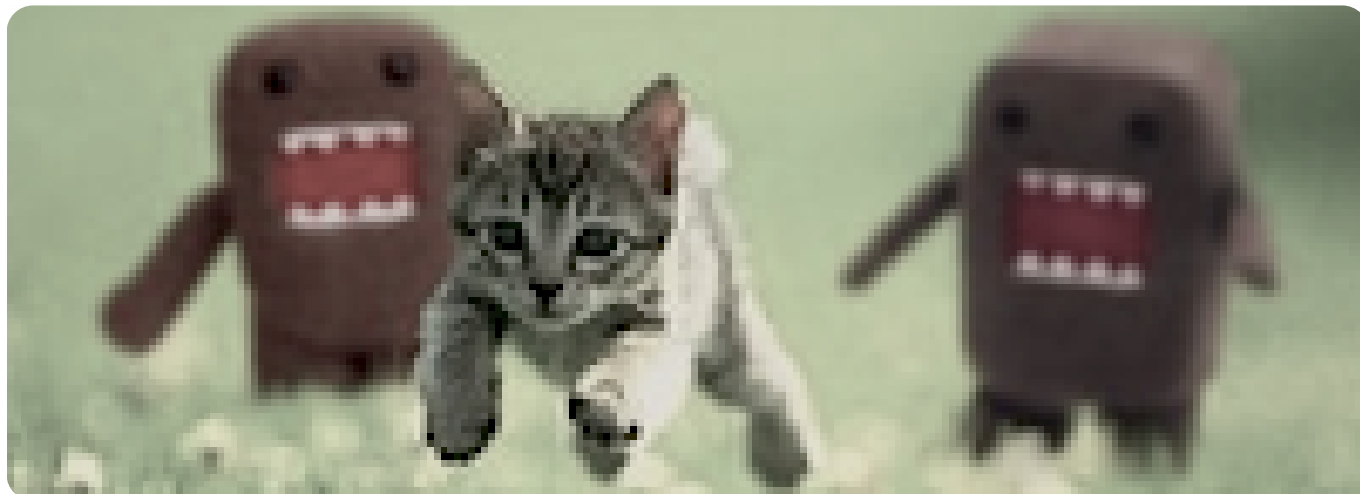


PREPARATION BOOTCAMP

GAME



If you need a little break in your day, we suggest you play [the bandit wargame](#). We also encourage you to resume your game later.



Conditionals



From now on, if ever you are stuck on a task, we warmly recommend trying a no-code tool, such as [Flowgorithm](#), that generates code from graphical flowcharts.

Task 1.1



Evaluate and explain the following lines:

- ✓ `(42 > 12)`
- ✓ `(12 = 12)`
- ✓ `(12 == 12)`
- ✓ `("hello" == "world")`
- ✓ `(218 >= 118)`
- ✓ `("a".upper() == "A")`
- ✓ `(1 * 2 * 3 * 4 <= 9)`
- ✓ `("z" in "azerty")`

Task 1.2



Ask the user for an integer. If it's equal to 42, display "This is correct!".

Task 1.3



Prompt the user for an integer. Then, if it is:

- ✓ odd, display "This integer is odd";
- ✓ even, display "This integer is even".

Task 1.4



Prompt the user for a string:

- ✓ if it's "open sesame", display "access granted";
- ✓ if it's "will you open, you goddamn !@&/°", display "access fucking granted";
- ✓ else, display "permission denied".

Task 1.5



Prompt the user for an integer:

- ✓ if it's 42, display "a";
- ✓ if it's smaller than or equal to 21, display "b";
- ✓ if it's even, display "c";

- ✓ if this integer divided by 2 is smaller than 21 (excluded), display "d";
- ✓ finally, if it is odd and greater than or equal to 45, display "e";
- ✓ in any other case, display "f".



For instance, the input 20 outputs bcd.

Task 1.6



Dig the following code. Then, run it. Finally, fix it.

```
a == 42
b == 41
if a = b
    print("A and B is the sames")
if b =< a
    print("B is equal or lower as A")
if b != a
    print("B his different from A")
```



While you are at it, also correct the grammar and orthographic mistakes...

what are other
words for
mistakes?



faults, errors, omission,
misstep, aberration, fault,
lapse, miscalculation, blunder,
inaccuracy



Loops

Task 2.1



Write a snippet of code that prints all integers from 1 to 1000.

Task 2.2



Prompt the user for a string. Display each character of this string twice.



For instance `taxi` will return `ttaxxi`.

Task 2.3



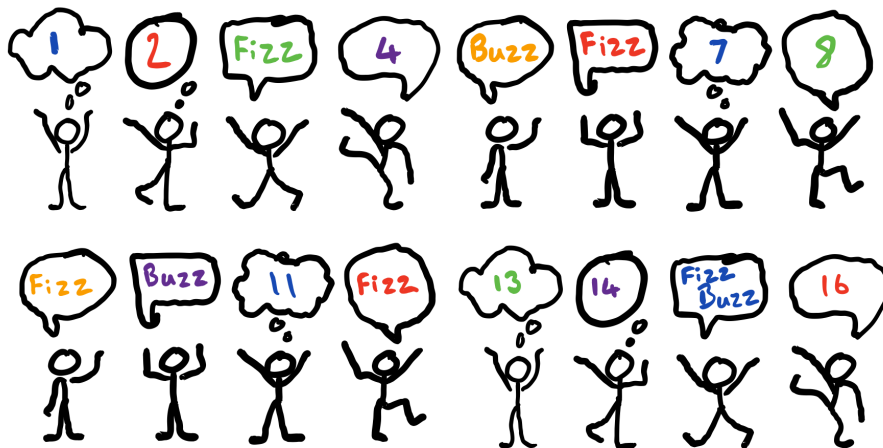
Print all integers, in decreasing order from 10 000 to 1, that are divisible by 7.

Task 2.4



For all integers from -30 to 30:

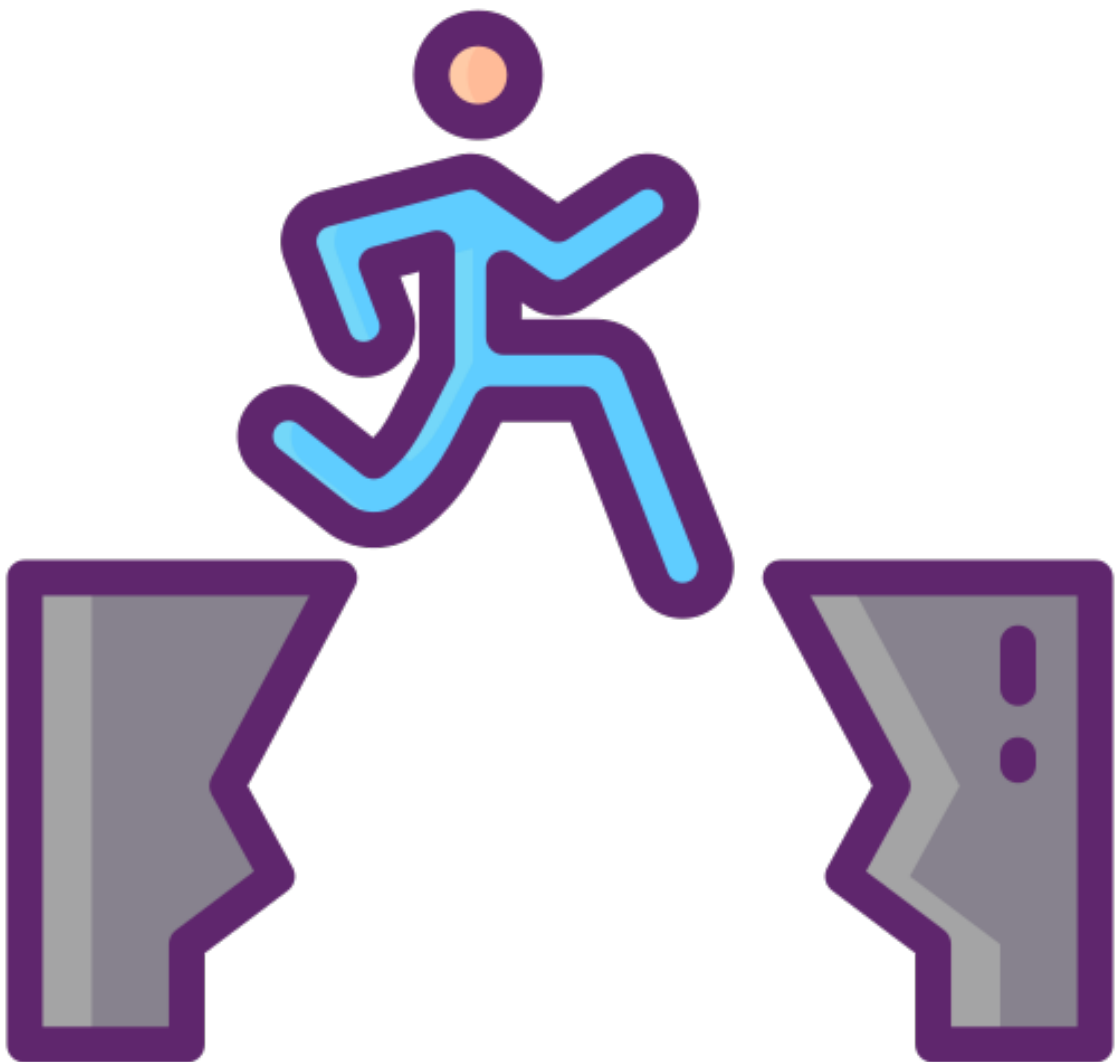
- ✓ if it's a multiple of 3, display "Fizz";
- ✓ if it's a multiple of 5, display "Buzz";
- ✓ if it's a multiple of 3 and 5, display "FizzBuzz";
- ✓ if it does not meet any of the previous conditions, just print the integer itself.



CHALLENGE

Write the shortest code possible that does the following:

- ✓ prompt the user simultaneously for an integer and a string;
- ✓ if the integer is 0, then quit;
- ✓ if the string contains a vowel, display the integer;
- ✓ if the integer is greater than or equal to 42, display the integer;
- ✓ else display the string.



Encryption

Task 3.1



The **Caesar Cipher** is a simple encryption technique named after Julius Caesar, who used it to encrypt messages of military significance.

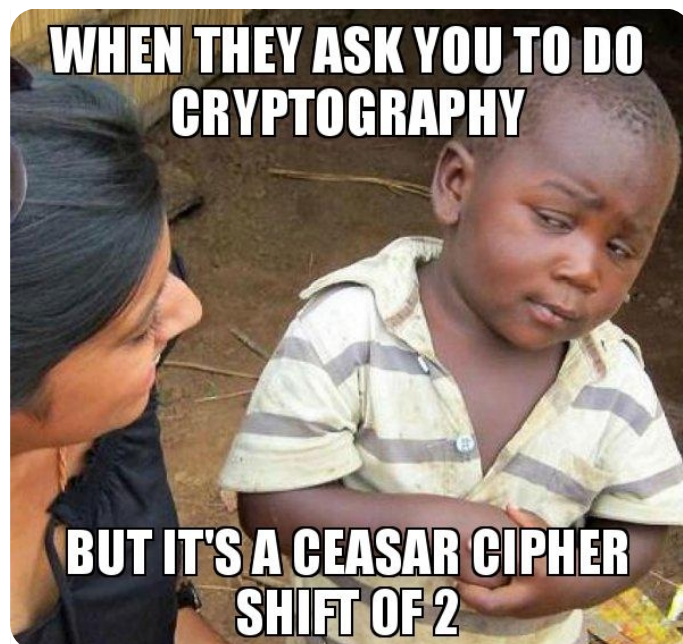
It is a type of substitution cipher in which each letter of a plaintext is replaced by a letter a fixed number of positions down the alphabet.

The encryption key is an integer between 1 and 25 that determines the shift of the letters. To decrypt the message, the recipient simply uses the same key to shift the letters back.

As examples:

- ✓ ab with a key of 3 gives de;
- ✓ hello world with a key of 4 gives lipps asvph;
- ✓ the zen of python with a key of 13 gives gur mra bs clguba.

Prompt the user for a clear message and a key. Then, use Caesar cipher to encrypt the clear message with the key. Finally, print the encrypted message.



Websites such as [dcode](#) may help you.

Is that not enough for you?



Task 3.2



Write a program that can decrypt any Caesar-ciphered text.

Task 3.3



Write a program that can encrypt and decrypt a text using a [Vigenère cipher](#).

Task 3.4



Write a program that can decrypt an English Vigenère-ciphered text given the length of the key, but without knowing the key.



Day 03 Task 3.5

v 2.2

{EPITECH}